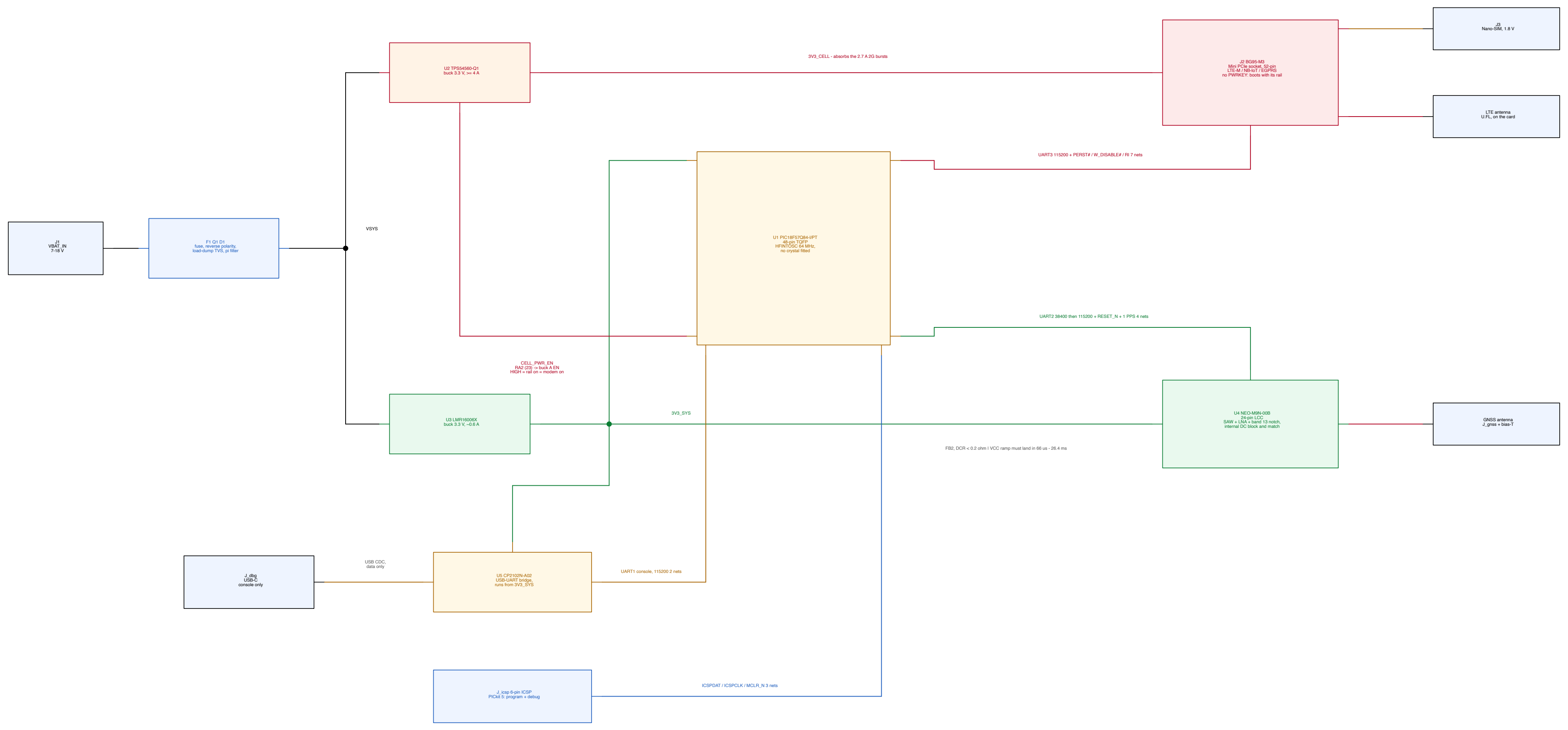
FARSEER - WHOLE-BOARD SCHEMATIC

In-car race telemetry module: PIC18F57Q84 + BG95-M3 Mini PCIe (UDP uplink / downlink) + NEO-M9N GNSS

Sheet 0 is the system overview and the cross-board net list. Sheets A-D are the four detail schematics, unchanged, each generated from its own script and matching its own BOM 1:1.

(0) SYSTEM OVERVIEW - every inter-block connection. Pin numbers: U1 = 48-pin TQFP, J2 = 52-pin Mini PCIe edge, U4 = 24-pin LCC. Wires that cross WITHOUT a dot are not connected.


J1
VBAT_IN
7-18 V
F1 Q1 D1
fuse, reverse polarity,
load-dump TVS, pi filter
VSYS
U2 TPS54560-Q1
buck 3.3 V, >= 4 A
3V3_CELL - absorbs the 2.7 A 2G bursts
J2 BG95-M3
Mini PCIe socket, 52-pin
LTE-M / NB-IoT / EGPRS
no PWRKEY: boots with its rail
J3
Nano-SIM, 1.8 V
LTE antenna
U.FL, on the card
UART3 115200 + PERST# / W_DISABLE# / RI 7 nets
U1 PIC18F57Q84-I/PT
48-pin TQFP
HFINTOSC 64 MHz,
no crystal fitted
CELL_PWR_EN
RA2 (23) -> buck A EN
HIGH = rail on = modem on
UART2 38400 then 115200 + RESET_N + 1 PPS 4 nets
U3 LMR16006X
buck 3.3 V, ~0.6 A
3V3_SYS
U4 NEO-M9N-00B
24-pin LCC
SAW + LNA + band 13 notch,
internal DC block and match
GNSS antenna
J_gnss + bias-T
FB2, DCR < 0.2 ohm / VCC ramp must land in 66 us - 26.4 ms
J_dbg
USB-C
console only
USB CDC,
data only
U5 CP2102N-A02
USB-UART bridge,
runs from 3V3_SYS
UART1 console, 115200 2 nets
J_icsp 6-pin ICSP
PICkit 5: program + debug
ICSPDAT / ICSPCLK / MCLR_N 3 nets

CROSS-BOARD NET LIST - every signal crossing a block boundary. No level shifters anywhere: the card's UART and control pins are 3.3 V, and the NEO shares the PIC's rail.

U1 <-> J2 BG95-M3 Mini PCIe socket

CELL_RX	RF4 (12) -> J2 pin 11 UART_RX	220 ohm series
CELL_TX	RF5 (13) <- J2 pin 13 UART_TX	220 ohm series
CELL_RTS	RF7 (15) -> J2 pin 23 UART_CTS	Opt
CELL_CTS	RF6 (14) <- J2 pin 25 UART_RTS	Opt
CELL_PERST	RB0 (8) -> J2 pin 22 PERST#	open-drain, 2-3.8 s
CELL_W_DIS	RB1 (9) -> J2 pin 20 W_DISABLE#	Opt
CELL_RI	RB2 (10) <- J2 pin 17 RI	Opt, INT2 wake

Card-relative names, ALREADY CROSSED: pin 11 is "connect to the DTE's TX". Do not cross the pair a second time.

U1 <-> U4 NEO-M9N

GNSS_RX	RD0 (42) -> U4 pin 21 RXD	
GNSS_TX	RD1 (43) <- U4 pin 20 TXD	
GNSS_RESET	RB3 (11) -> U4 pin 8 RESET_N	open-drain
GNSS_1PPS	RC2 (40) <- U4 pin 3 TIMEPULSE	CCP1, 30 ns RMS

No hardware flow control exists on the NEO-M9N. TIMEPULSE is routed even though the firmware does not use it yet: hardware capture beats UART arrival time by orders of magnitude for tying a position fix to lap timing.

U1 <-> U5 / J_icsp console and programming

DBG_TX	RF0 (36) -> U5 RXD (20)	220 ohm series
DBG_RX	RF1 (37) <- U5 TXD (21)	220 ohm series
ICSPDAT	RB7 (19) <-> J_icsp pin 4	
ICSPCLK	RB6 (18) <-> J_icsp pin 5	
MCLR_N	RE3 (20) <- J_icsp pin 1 VPP	
CELL_PWR_EN	RA2 (23) -> buck A EN	HIGH = rail on

U5 TXD already drives the PIC RX - do not cross again. Programming and in-circuit debug are J_icsp with a PICkit 5; the PIC18F57Q84 has no USB peripheral, so USB-C is console only.

RAILS. VBAT_IN 7-18 V -> VSYS -> two independent bucks. 3V3_CELL absorbs the BG95's 2.7 A 2G bursts; 3V3_SYS stays quiet for the PIC, the NEO-M9N and the USB bridge, so a transmit burst cannot brown out the MCU or the GNSS receiver. Both rails are 3.3 V and must stay 3.3 V - raising either one breaks cross-rail logic levels. Single ground plane throughout.

BOOT ORDER. Buck B is always enabled, so the PIC and the NEO-M9N come up together and can never drive each other while one is unpowered. Buck A's EN pull-down holds the modem off while the PIC is in reset; the PIC then drives CELL_PWR_EN high and the card auto-powers on - there is no PWRKEY on this card. PERST# is for recovering a wedged modem, not for boot.

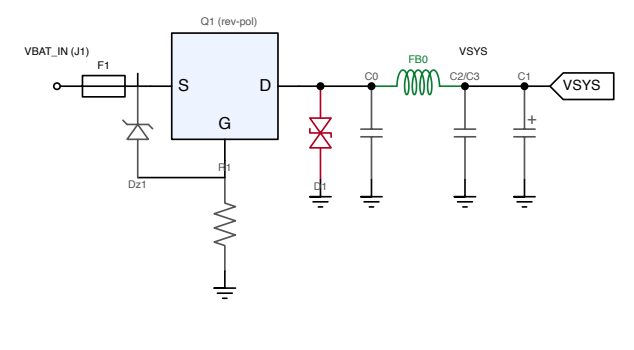
READING SHEETS A-D. Within them, connections are made by NET-LABEL TAG rather than by long wires, and the same tag name on two tiles is the same net: 3V3_SYS on sheet A is the same copper as 3V3_SYS on sheets B and D. Reference designators only are printed; component values live in the matching BOM markdown. Pin budget and PPS checks: interconnect-and-pin-budget.md.

(A) POWER RAILS component values in power-supply-bom.md

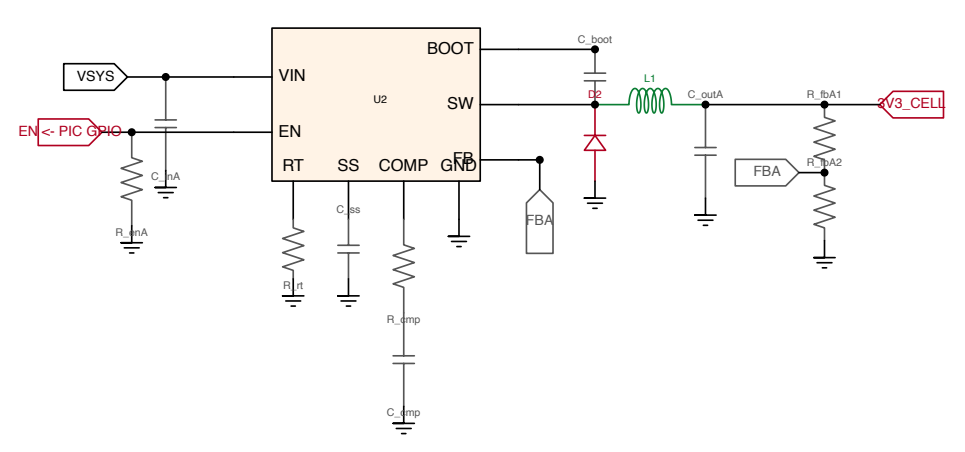
FARSEER POWER RAILS - detailed schematic

Reference designators shown; component values are in power-supply-bom.md

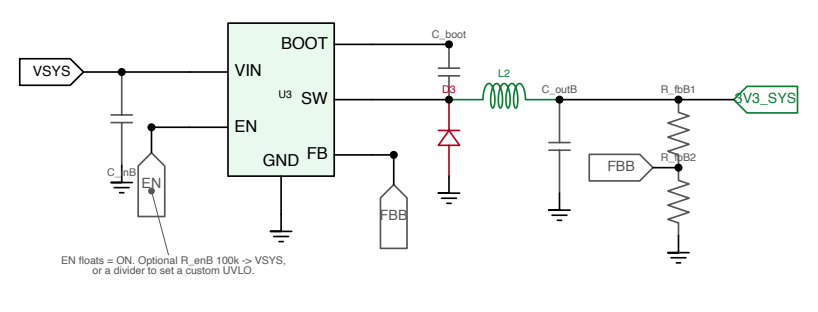
(1) VEHICLE INPUT -> PROTECTION -> VSYS


Q1 (rev-pol)
VBAT_IN (J1)
F1
S
D
G
Dz1
R1
D1
C0
FB0
VSYS
C2/C3
C1
VSYS

(2) BUCK A U2 TPS54560-Q1 : VSYS -> 3.3 V (>=4 A, 2G-capable) -> 3V3_CELL

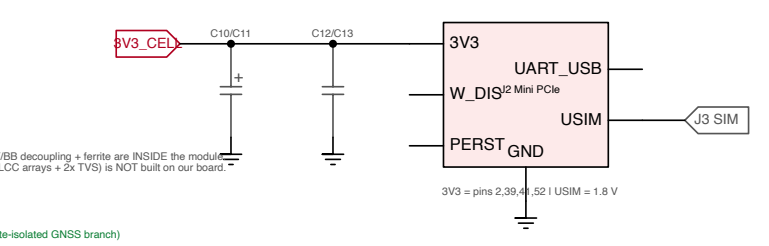

VSYS
EN <- PIC GPIO
VIN
EN
RT
SS
COMP
GND
BOOT
SW
FB
U2
C_inA
R_enA
R_rt
C_ss
R_cmp
C_cmp
C_boot
D2
L1
C_outA
R_fbA1
R_fbA2
FBA
FBA
3V3_CELL

(3) BUCK B U3 LMR16006X : VSYS -> 3.3 V (~0.6 A) -> 3V3_SYS (comp + soft-start internal)


VSYS
VIN
EN
GND
BOOT
SW
FB
U3
C_inB
EN
C_boot
D3
L2
C_outB
R_fbB1
R_fbB2
FBB
FBB
3V3_SYS

EN floats = ON. Optional R_enB 100k -> VSYS, or a divider to set a custom UVLO.

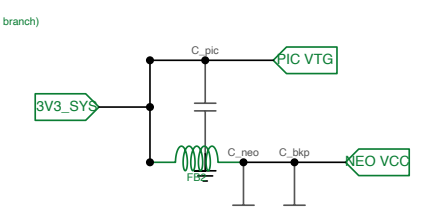
(4) 3V3_CELL -> BG95-M3 MINI PCIe SOCKET


3V3_CELL
C10/C11
C12/C13
3V3
W_DIS
PERST
UART_USB
USIM
GND
J2 Mini PCIe
J3 SIM

Card is self-contained: on-card VBAT regulation, RF/BB decoupling + ferrite are INSIDE the module. The raw-module VBAT star network (ferrite + MLCC arrays + 2x TVS) is NOT built on our board.

3V3 = pins 2,39,41,52 | USIM = 1.8 V

(5) 3V3_SYS -> PIC VTG + NEO-M9N (ferrite-isolated GNSS branch)


C_pic
PIC VTG
3V3_SYS
FB2
C_neo
C_bkp
NEO VCC

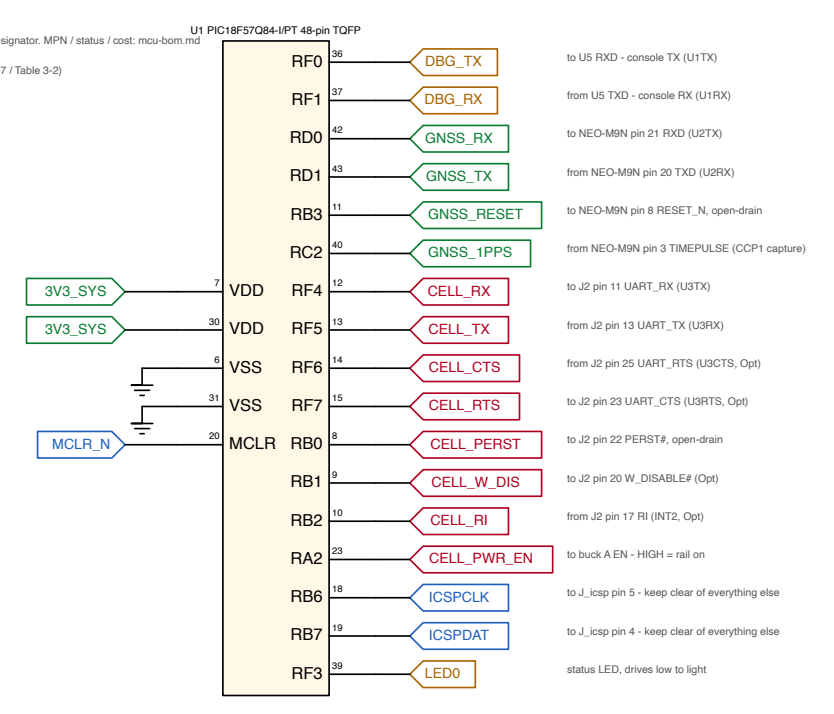
(B) MCU SUPPORT component values in mcu-support-circuit.md

FARSEER MCU SUPPORT CIRCUIT - PIC18F57Q84, 48-pin TQFP

Every resistor and capacitor shows its VALUE next to the reference designator. MPN / status / cost: mcu-bom.md

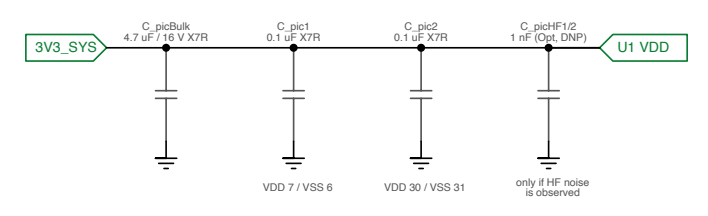
All pin numbers are the 48-pin TQFP column (DS40002213F Figure 2-7 / Table 3-2)

(1) U1 PIC18F57Q84-I/PT : every connected pin


U1 PIC18F57Q84-I/PT 48-pin TQFP
3V3_SYS
7 VDD
3V3_SYS
30 VDD
6 VSS
31 VSS
MCLR_N
20 MCLR
RF0 36 DBG_TX to U5 RXD - console TX (U1TX)
RF1 37 DBG_RX from U5 TXD - console RX (U1RX)
RD0 42 GNSS_RX to NEO-M9N pin 21 RXD (U2TX)
RD1 43 GNSS_TX from NEO-M9N pin 20 TXD (U2RX)
RB3 11 GNSS_RESET to NEO-M9N pin 8 RESET_N, open-drain
RC2 40 GNSS_1PPS from NEO-M9N pin 3 TIMEPULSE (CCP1 capture)
RF4 12 CELL_RX to J2 pin 11 UART_RX (U3TX)
RF5 13 CELL_TX from J2 pin 13 UART_TX (U3RX)
RF6 14 CELL_CTS from J2 pin 25 UART_RTS (U3CTS, Opt)
RF7 15 CELL_RTS to J2 pin 23 UART_CTS (U3RTS, Opt)
RB0 8 CELL_PERST to J2 pin 22 PERST#, open-drain
RB1 9 CELL_W_DIS to J2 pin 20 W_DISABLE# (Opt)
RB2 10 CELL_RI from J2 pin 17 RI (INT2, Opt)
RA2 23 CELL_PWR_EN to buck A EN - HIGH = rail on
RB6 18 ICSPCLK to J_icsp pin 5 - keep clear of everything else
RB7 19 ICSPDAT to J_icsp pin 4 - keep clear of everything else
RF3 39 LED0 status LED, drives low to light

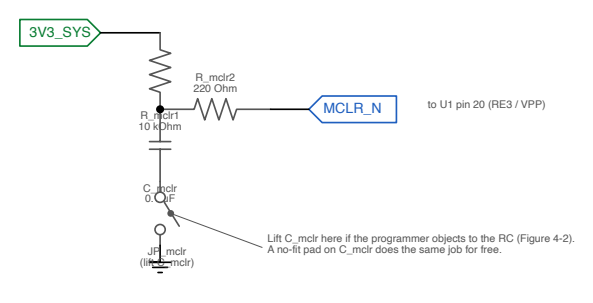
No AVDD / AVSS and no VCAP on this device, so ADC quality depends entirely on how clean 3V3_SYS is at pins 7 and 30 - there is no analog supply to filter separately, and VREF is not a dedicated pin (it would cost RA2 / RA3). Unused I/O: drive low in firmware (DS40002213F 4.6) or tie to VSS through 1-10 kOhm.

(2) SUPPLY AND DECOUPLING : 0.1 uF per VDD pin, REQUIRED (DS40002213F 4.2.1)


3V3_SYS
C_picBulk
4.7 uF / 16 V X7R
C_pic1
0.1 uF X7R
C_pic2
0.1 uF X7R
C_picHF1/2
1 nF (Opt, DNP)
U1 VDD
VDD 7 / VSS 6
VDD 30 / VSS 31
only if HF noise
is observed

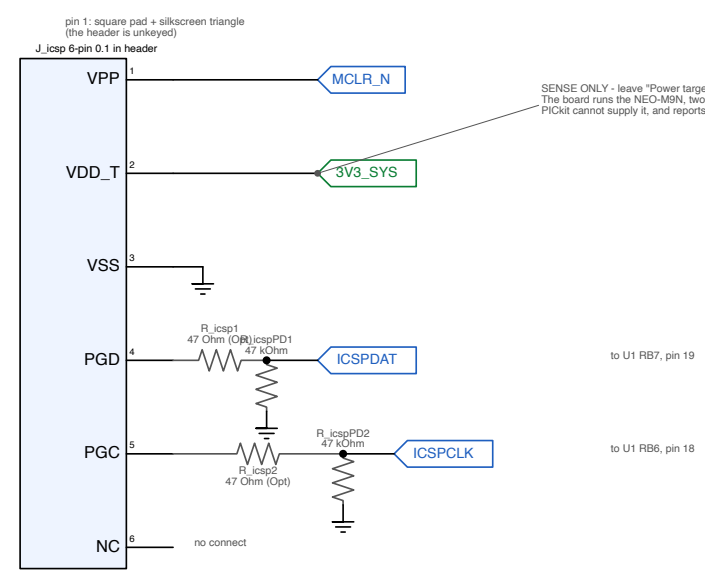
Pins 7/6 and 30/31 sit on OPPOSITE CORNERS of the package, so these are two genuinely separate placements - one shared cap is not compliant. Each cap: ceramic, self-resonant at 200 MHz or above, same side of the board as the MCU, trace no longer than 6 mm, and power routed to the CAP FIRST and then on to the device pin.

(3) MCLR RESET NETWORK (DS40002213F Figure 4-1)


3V3_SYS
R_mclr2
220 Ohm
MCLR_N
to U1 pin 20 (RE3 / VPP)
R_mclr1
10 kOhm
C_mclr
0.1 uF
JP_mclr
(lift C_mclr)
Lift C_mclr here if the programmer objects to the RC (Figure 4-2).
A no-fit pad on C_mclr does the same job for free.

All three parts within 6 mm of pin 20 (4.3). R_mclr1 <= 10 kOhm and R_mclr2 <= 470 Ohm are datasheet limits, not suggestions: R_mclr2 caps the current C_mclr can dump into MCLR during an ESD/EOS breakdown. The Curiosity Nano fits 47 kOhm and no capacitor, which violates R1 <= 10 kOhm - follow the datasheet, not the dev board.

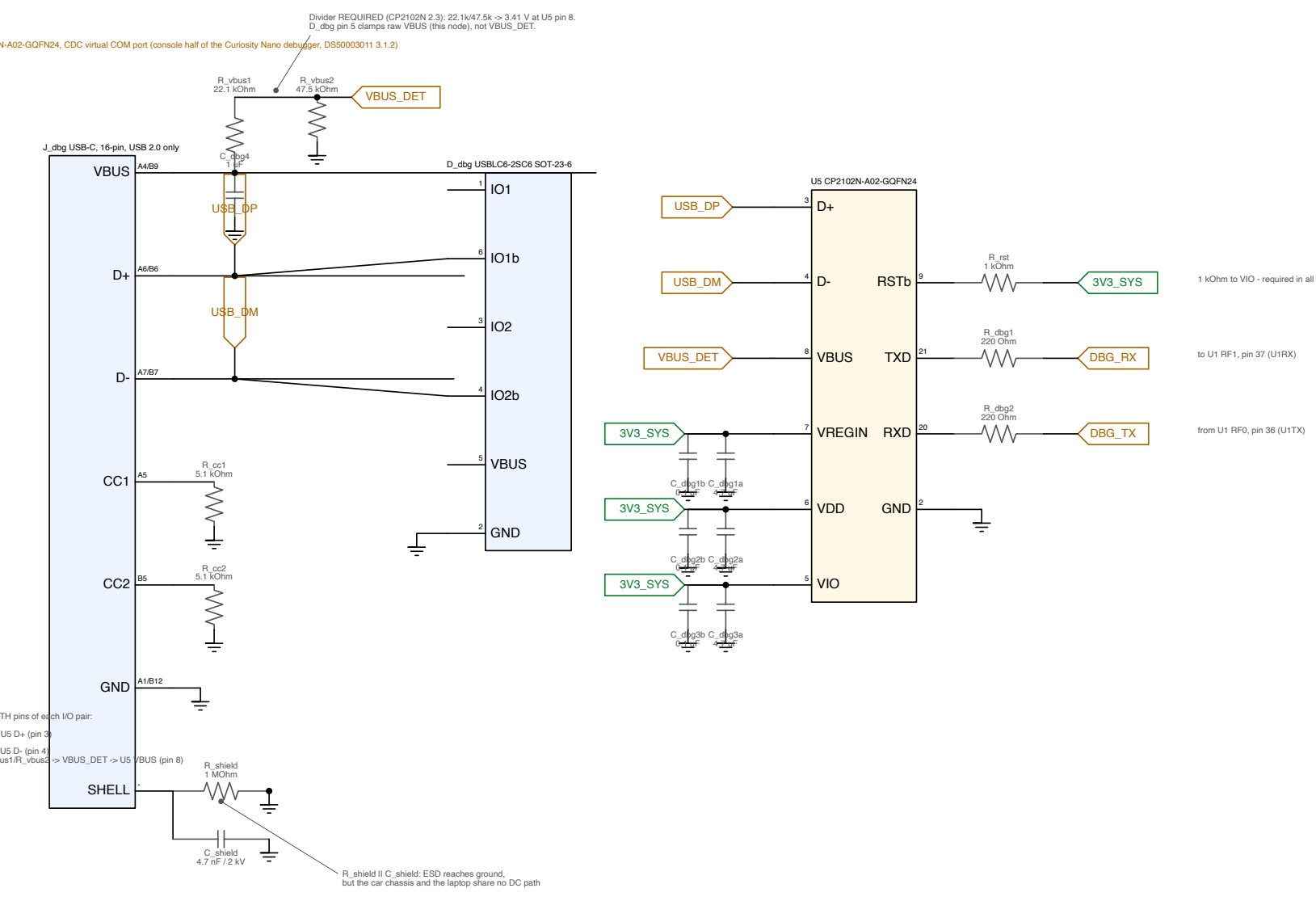
(4) ICSP PROGRAM + DEBUG HEADER - PICkit 5 (Microchip PICkit pinout, Figures 48-1 / 48-2)


pin 1: square pad + silkscreen triangle
(the header is unkeyed)
J_icsp 6-pin 0.1 in header
1 VPP
MCLR_N
2 VDD_T
3V3_SYS
SENSE ONLY - leave "Power target circuit from tool" OFF in MPLAB. The board runs the NEO-M9N, two bucks and a 2.7 A BG95 card; the PICkit cannot supply it, and reports a target-voltage error if asked.
3 VSS
4 PGD
R_icsp1
47 Ohm (Opt)
R_icspPD1
47 kOhm
ICSPDAT
to U1 RB7, pin 19
5 PGC
R_icsp2
47 Ohm (Opt)
R_icspPD2
47 kOhm
ICSPCLK
to U1 RB6, pin 18
6 NC
no connect

DS40002213F 4.4 is emphatic about what NOT to add: no pull-UPS, series diodes, or capacitors on ICSPCLK / ICSPDAT - they interfere with programmer communications. R_icsp1/2 are ESD series only and must not exceed 100 Ohm; the 47 kOhm pull-DOWNS are what Microchip debug tooling expects. Nothing else may share RB6 / RB7.

THIS HEADER IS BOTH PROGRAMMER AND DEBUGGER, permanently, not a manufacturing fixture. The device offers In-Circuit Debug with three breakpoints over the SAME two pins (DS40002213F p. 6); MPLAB clears DEBUG (CONFIG7[5]) for a debug session and no board change is needed. JTAG is not an alternative - 39 states the PIC18-Q84 JTAG module does not support programming. Leave LVP (CONFIG4[5]) at its erased 1: clearing it forces high-voltage VPP entry from then on, and only HV programming can ever set it back.

(5) DEBUG CONSOLE - USB-C to CP2102N-A02-GQFN24, CDC virtual COM port (console half of the Curiosity Nano debugger, DS50003011 3.1.2)

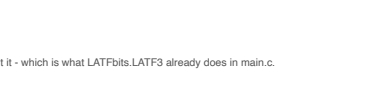

Divider REQUIRED (CP2102N 2.3): 22.1k/47.5k -> 3.41 V at U5 pin 8.
D_dbg pin 5 clamps raw VBUS (this node), not VBUS_DET.
R_vbus1
22.1 kOhm
R_vbus2
47.5 kOhm
VBUS_DET
J_dbg USB-C, 16-pin, USB 2.0 only
VBUS A4/B9
C_dbg4
1 uF
USB_DP
D+ A6/B6
USB_DM
D- A7/B7
CC1 A5
R_cc1
5.1 kOhm
CC2 B5
R_cc2
5.1 kOhm
GND A1/B12
SHELL
R_shield
1 MOhm
C_shield
4.7 nF / 2 kV
R_shield II C_shield: ESD reaches ground,
but the car chassis and the laptop share no DC path
D_dbg USBLC6-2SC6 SOT-23-6
1 IO1
6 IO1b
3 IO2
4 IO2b
5 VBUS
2 GND
U5 CP2102N-A02-GQFN24
USB_DP 3 D+
USB_DM 4 D-
VBUS_DET 8 VBUS
3V3_SYS 7 VREGIN
3V3_SYS 6 VDD
3V3_SYS 5 VIO
C_dbg1b C_dbg1a
C_dbg2b C_dbg2a
C_dbg3b C_dbg3a
RSTb 9
R_rst
1 kOhm
3V3_SYS
1 kOhm to VIO - required in all cases
TXD 21
R_dbg1
220 Ohm
DBG_RX
to U1 RF1, pin 37 (U1RX)
RXD 20
R_dbg2
220 Ohm
DBG_TX
from U1 RF0, pin 36 (U1TX)
GND 2

D_dbg (USBLC6-2SC6) pin hookup - wire BOTH pins of each I/O pair:
pin 1 I/O1 -+
pin 6 I/O1 -+- USB_DP = J_dbg D+ (A6/B6) = U5 D+ (pin 3)
pin 3 I/O2 -+
pin 4 I/O2 -+- USB_DM = J_dbg D- (A7/B7) = U5 D- (pin 4)
pin 5 VBUS ---- J_dbg VBUS (A4/B9) -> R_vbus1/R_vbus2 -> VBUS_DET -> U5 VBUS (pin 8)
pin 2 GND ---- GND

U5 runs from 3V3_SYS, NOT from VBUS. This is CP2102N Figure 2.3 (internal 5 V regulator unused, so VREGIN is tied to VDD) plus Figure 2.6 (self-powered USB, VBUS reaching the SENSE PIN ONLY). The bus-powered Figure 2.5 ties VBUS to VREGIN instead - that is the wiring to avoid, because it would let a laptop run the bridge while the vehicle rail is dead and drive DBG_RX into an unpowered MCU. The back-power failure mode from bench-wiring.md is designed out rather than patched. VBUS only ever tells U5 that a host is attached.

THE CROSS IS ALREADY DONE: U5 TXD (21) drives the PIC RX (RF1) and U5 RXD (20) listens to the PIC TX (RF0), exactly as the Nano wires CDC TX to the target UART RX (DS50003011 Figure 3-1). Do not cross it a second time. Both directions must be populated: RF1 is how a server URL will later be written into the 1 KB Data EEPROM, so an output-only console would foreclose it. No DTR/RTS auto-reset circuit - reset belongs to J_icsp, and an RC on MCLR outside the sheet-3 network would fight the PICkit for the pin. Two SEPARATE 5.1 kOhm CC resistors are required, and D+/D- must reach BOTH A6/A7 and B6/B7 for either cable orientation to work. Route the pair at 90 Ohm differential. No series resistors on D+/D-: the CP2102N integrates the transceiver, its matching, and the pull-ups. Programming is NOT here - the PIC18F57Q84 has no USB peripheral, so U5 is the virtual COM port only. Program and debug are on J_icsp with a PICkit 5.

(6) STATUS LED - RF3, pin 39, active low


D_led0
~2 mA
R_led0
1 kOhm
3V3_SYS
LED0
to U1 RF3, pin 39

RF3 sinks the LED current, so the pin drives LOW to light it - which is what LATFbits.LATF3 already does in main.c.

(7) CLOCK - nothing fitted

HFINTOSC at 64 MHz. Y_hs, C_osc1, C_osc2 and Y_sosc are all quantity ZERO, which leaves OSC1 (32) and OSC2 (33) free as general I/O. CONFIG1 MUST be programmed with RSTOSC = 000: an erased part comes up with RSTOSC = 111, meaning "external clock", and a crystal-less board will simply never run.

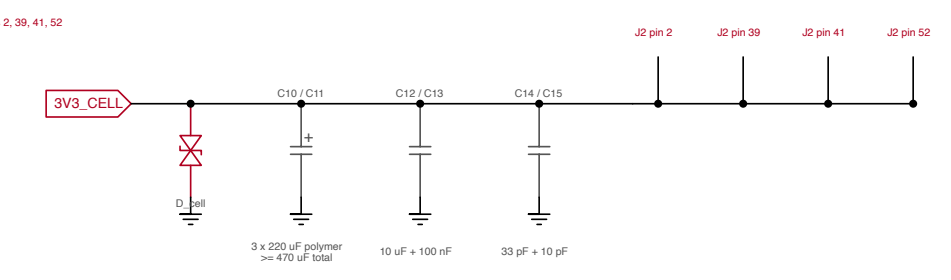
(C) CELLULAR - BG95-M3 component values in cellular-support-circuit.md

FARSEER CELLULAR SUPPORT CIRCUIT - Quectel BG95-M3, Mini PCIe card

Reference designators shown; component values are in cellular-support-circuit.md

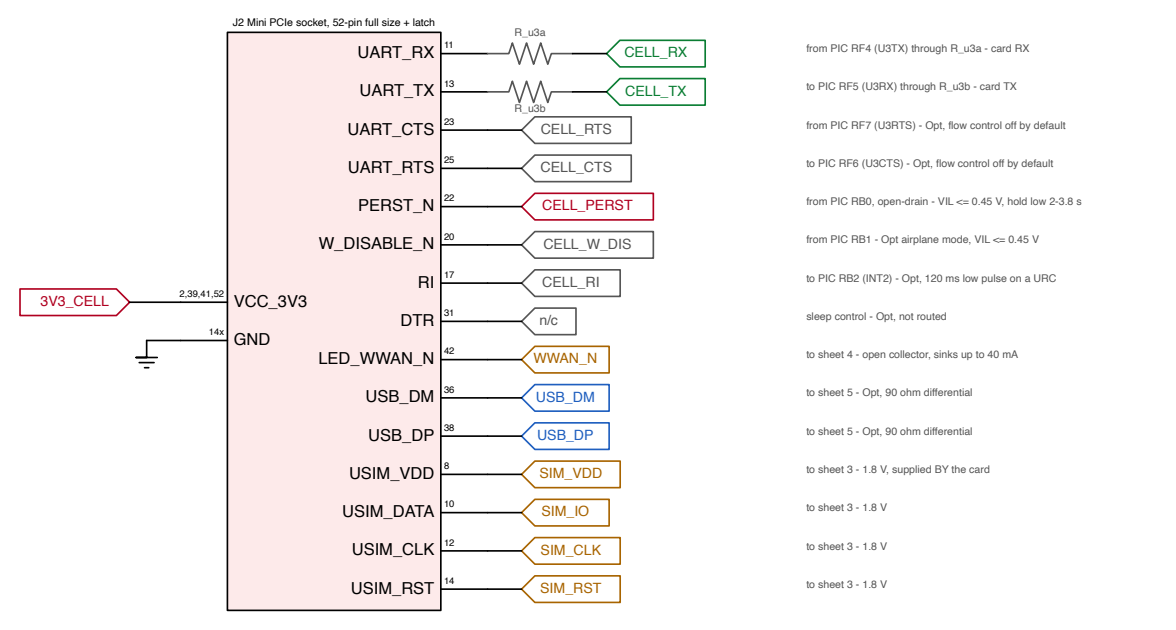
Pin numbers are the 52-pin Mini PCIe edge connector (Mini PCIe Hardware Design V1.0, Table 4)

(1) SOCKET POWER : 3V3_CELL -> VCC_3V3 pins 2, 39, 41, 52


J2 pin 2
J2 pin 39
J2 pin 41
J2 pin 52
3V3_CELL
D_cell
C10 / C11
C12 / C13
C14 / C15
3 x 220 uF polymer
>= 470 uF total
10 uF + 100 nF
33 pF + 10 pF

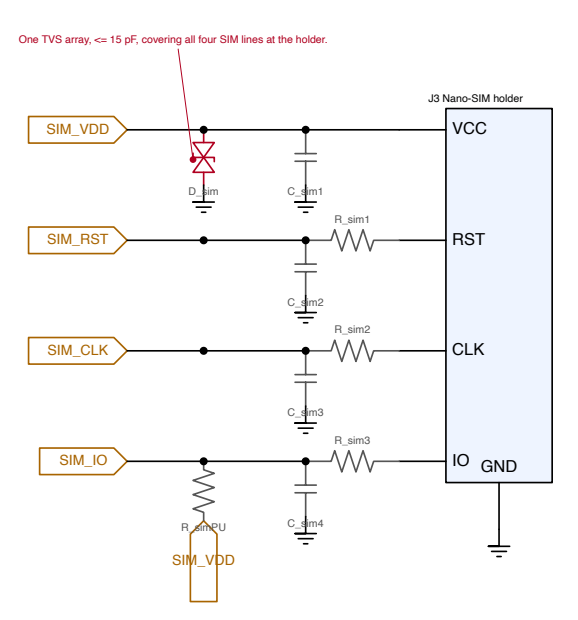
Quectel 3.2: in 2G the input peak current reaches 2.7 A, and a LOW-ESR BYPASS CAPACITOR OF NO LESS THAN 470 uF is required to keep the rail from collapsing. C10/C11 is that reservoir; C12-C15 complete the Figure 3 ladder. All four VCC_3V3 pins must be fed - they are spread across the connector deliberately, and feeding only one is a current-density mistake. All fourteen GND pins (4, 9, 15, 18, 21, 26, 27, 29, 34, 35, 37, 40, 43, 50) go to plane. Sizing arithmetic is in power-rail-notes.md. Keep buck A's inductor and switch node away from the antenna run.

(2) J2 SIGNAL MAP : every non-power pin we touch


J2 Mini PCIe socket, 52-pin full size + latch
3V3_CELL
2,39,41,52 VCC_3V3
14x GND
UART_RX 11 R_u3a CELL_RX from PIC RF4 (U3TX) through R_u3a - card RX
UART_TX 13 R_u3b CELL_TX to PIC RF5 (U3RX) through R_u3b - card TX
UART_CTS 23 CELL_RTS from PIC RF7 (U3RTS) - Opt, flow control off by default
UART_RTS 25 CELL_CTS to PIC RF6 (U3CTS) - Opt, flow control off by default
PERST_N 22 CELL_PERST from PIC RB0, open-drain - VIL <= 0.45 V, hold low 2-3.8 s
W_DISABLE_N 20 CELL_W_DIS from PIC RB1 - Opt airplane mode, VIL <= 0.45 V
RI 17 CELL_RI to PIC RB2 (INT2) - Opt, 120 ms low pulse on a URC
DTR 31 n/c sleep control - Opt, not routed
LED_WWAN_N 42 WWAN_N to sheet 4 - open collector, sinks up to 40 mA
USB_DM 36 USB_DM to sheet 5 - Opt, 90 ohm differential
USB_DP 38 USB_DP to sheet 5 - Opt, 90 ohm differential
USIM_VDD 8 SIM_VDD to sheet 3 - 1.8 V, supplied BY the card
USIM_DATA 10 SIM_IO to sheet 3 - 1.8 V
USIM_CLK 12 SIM_CLK to sheet 3 - 1.8 V
USIM_RST 14 SIM_RST to sheet 3 - 1.8 V

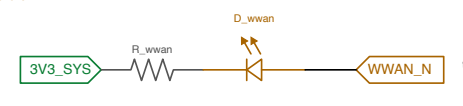
NO LEVEL SHIFTERS. Table 8 puts UART_RX/TX/CTS/RTS in the 3.3 V power domain and Table 10 does the same for RI, DTR, W_DISABLE# and PERST#, so the 3.3 V PIC wires straight across. Do not fit the TXS0108 the raw-LGA guide calls for. Only (U)SIM, PCM and I2C are 1.8 V, and none of those reach the MCU.
The UART names are CARD-RELATIVE and therefore ALREADY CROSSED: pin 11 UART_RX is "connect to DTE's TX" and pin 13 UART_TX is "connect to DTE's RX". Wire PIC TX to pin 11 and PIC RX to pin 13; do not cross them a second time.
There is NO PWRKEY on this card. Figure 1 shows an automatic power-on circuit, so the card boots whenever 3V3_CELL is live - boot order is set by buck A's EN pin (PIC RA2, high = on), not by a module key pin.
LEFT UNCONNECTED: WAKE# (1), I2C_SCL/SDA (30, 32), USIM_DET (44), PCM_CLK/DOUT/DIN/SYNC (45, 47, 49, 51), and every RESERVED pin (3, 5, 6, 7, 16, 19, 24, 28, 33, 46, 48). Table 4 note 3: keep all reserved and unused pins unconnected - do NOT ground them "to be safe", since several carry repurposed standard Mini PCIe functions.

(3) (U)SIM HOLDER - 1.8 V only, Quectel Figure 5 (6-pin holder)


One TVS array, <= 15 pF, covering all four SIM lines at the holder.
J3 Nano-SIM holder
SIM_VDD
D_sim
C_sim1
VCC
SIM_RST
R_sim1
C_sim2
RST
SIM_CLK
R_sim2
C_sim3
CLK
SIM_IO
R_simPU
SIM_VDD
R_sim3
C_sim4
IO
GND

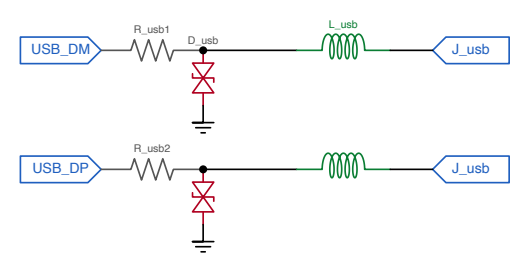
The card supplies USIM_VDD itself at 1.8 V; our board provides the holder and the protection network. C_sim1 must NOT exceed 1 uF (3.3) and sits at the holder. The 33 pF caps filter EGSM900 interference; the 0 ohm links are the debug aid Figure 5 calls for.
Layout, all mandatory at review: holder as close to the socket as possible with trace length under 200 mm; SIM signals kept away from RF and power-supply traces; ground between socket and holder short and wide, >= 0.5 mm; USIM_DATA and USIM_CLK separated and shielded by surrounding ground. The holder is the only user-accessible connector on this subsystem, so it is the realistic ESD entry point - D_sim is Req, not Rec.

(4) LED_WWAN# - network status, pin 42, open collector, active low


D_wwan
R_wwan
3V3_SYS
WWAN_N
to J2 pin 42

R_wwan IS NOT OPTIONAL if D_wwan is fitted: 3.7.4 says a resistor MUST be placed in series, and the pin will sink up to 40 mA and cook the LED without it. 1 kOhm gives about 2 mA. Default pattern (AT+QCFG="ledmode",0): 200 ms low / 1800 ms high = searching, 1800 / 200 = registered and idle, 125 / 125 = data transfer - readable across a paddock.

(5) USB - pins 36 / 38, optional, firmware update path


USB_DM
R_usb1
D_usb
L_usb
J_usb
USB_DP
R_usb2
J_usb

R_usb1/2 are 0 ohm and NOT MOUNTED by default (Figure 6) - they exist so the pair can be broken out to test points. L_usb is a common-mode choke and belongs close to the socket. Route as a 90 ohm differential pair with ground surround, never under crystals, oscillators, magnetics or RF traces, preferably on an inner layer.

(6) RF AND MECHANICAL - no host-board parts
RF: NOTHING on our board. The card carries its own 50 ohm main antenna connector and the antenna mates directly to it - no 50 ohm trace, no matching network, no antenna feed on the host PCB. What the board must provide is mechanical: clearance above the card for the U.FL-LP plug and its cable bend radius, a route to the enclosure bulkhead, and strain relief. In a race car that cable sees vibration continuously - anchor it.
Mechanical: 51.0 x 30.0 mm full-size card, standard 52-pin Mini PCIe socket (e.g. Molex 679105700) WITH latch. In a vehicle the latch is structural, not convenience; add a retaining screw or bracket. Operating range tops out at +75 C (extended +80 C), which an enclosure near a transmission tunnel can exceed - plan the mounting spot.

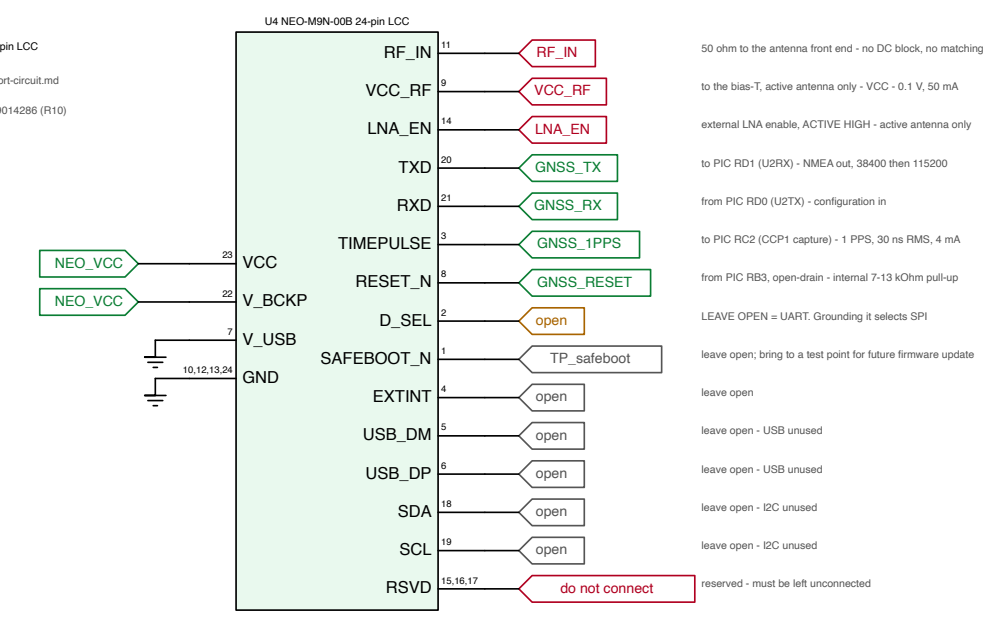
(D) GNSS - NEO-M9N component values in gnss-support-circuit.md

FARSEER GNSS SUPPORT CIRCUIT - u-blox NEO-M9N-00B, 24-pin LCC

Reference designators shown; component values are in gnss-support-circuit.md

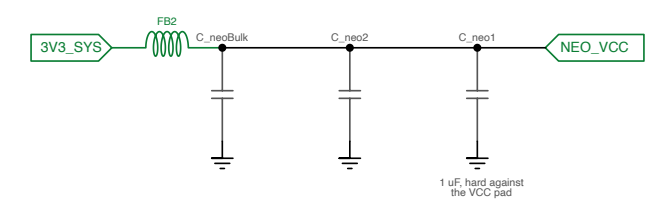
Sources: datasheet UBX-19014285 and integration manual UBX-19014286 (R10)

(1) U4 NEO-M9N-00B : all 24 pins, with the board action for each


U4 NEO-M9N-00B 24-pin LCC
NEO_VCC 23 VCC
NEO_VCC 22 V_BCKP
7 V_USB
10,12,13,24 GND
RF_IN 11 RF_IN 50 ohm to the antenna front end - no DC block, no matching
VCC_RF 9 VCC_RF to the bias-T, active antenna only - VCC - 0.1 V, 50 mA
LNA_EN 14 LNA_EN external LNA enable, ACTIVE HIGH - active antenna only
TXD 20 GNSS_TX to PIC RD1 (U2RX) - NMEA out, 38400 then 115200
RXD 21 GNSS_RX from PIC RD0 (U2TX) - configuration in
TIMEPULSE 3 GNSS_1PPS to PIC RC2 (CCP1 capture) - 1 PPS, 30 ns RMS, 4 mA
RESET_N 8 GNSS_RESET from PIC RB3, open-drain - internal 7-13 kOhm pull-up
D_SEL 2 open LEAVE OPEN = UART. Grounding it selects SPI
SAFEBOOT_N 1 TP_safeboot leave open; bring to a test point for future firmware update
EXTINT 4 open leave open
USB_DM 5 open leave open - USB unused
USB_DP 6 open leave open - USB unused
SDA 18 open leave open - I2C unused
SCL 19 open leave open - I2C unused
RSVD 15,16,17 do not connect reserved - must be left unconnected

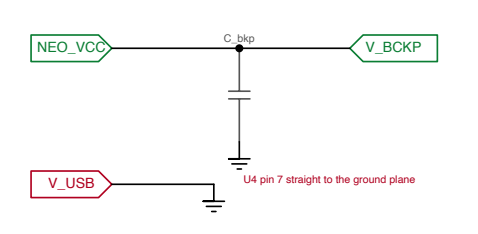
Baseline rule (datasheet 5): all inputs have internal pull-ups and can be left open if unused - which is why most of this pin list is "leave open" rather than a resistor. The two exceptions are V_USB, which must be GROUNDED, and V_BCKP, which must be TIED TO VCC if no backup source is fitted. Neither is covered by the leave-open rule.
The LCC package has perimeter castellations at 1.1 mm pitch: joints are visible and probeable, no thermal pad, no X-ray needed. Take the footprint and paste mask from IM Figures 37 and 38, and leave keep-out on the long edge for de-paneling residual tabs up to 0.5 mm.

(2) MAIN SUPPLY : 3V3_SYS -> FB2 -> NEO_VCC (pin 23)


3V3_SYS
FB2
C_neoBulk
C_neo2
C_neo1
NEO_VCC
1 uF, hard against
the VCC pad

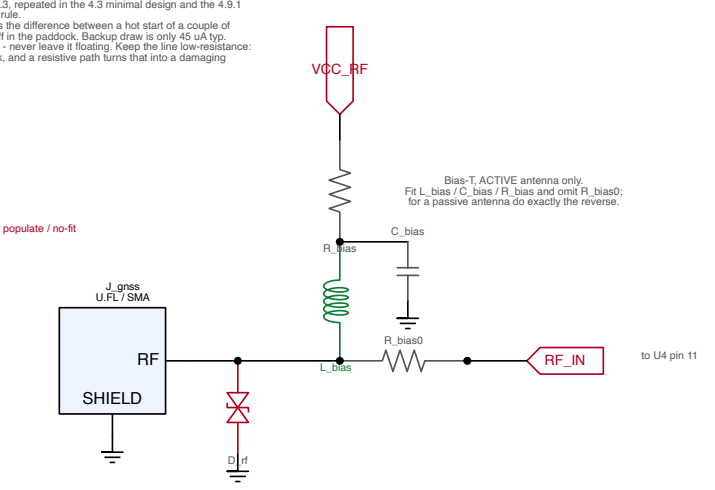
THREE CONSTRAINTS THAT ARE EASY TO VIOLATE.
1. Series resistance in the VCC line must stay UNDER 0.2 OHM (IM 4.2.1 / 4.9.1), or dynamic current generates input voltage noise. That is a spec on FB2: choose the bead for DCR below 0.2 ohm, not just for impedance at 100 MHz. A typical 600 ohm @ 100 MHz part can easily exceed it.
2. VCC ramp rate must land between 20 and 8000 us/V, so a 0 -> 3.3 V rise takes between 66 us and 26.4 ms. Too fast MAY PERMANENTLY DAMAGE THE DEVICE (datasheet Table 10 note 7). Buck B's ramp is checked in power-rail-notes.md, and must be re-checked if its soft-start ever changes.
3. The VCC pad wants copper, not a thin trace: large pad, large vias down to the power layer, C_neo1 as close as physically possible (IM 4.8.4.2). Peak draw is 100 mA during acquisition.

(3) V_BCKP (pin 22) AND V_USB (pin 7)


NEO_VCC
C_bkp
V_BCKP
to U4 pin 22
V_USB
U4 pin 7 straight to the ground plane

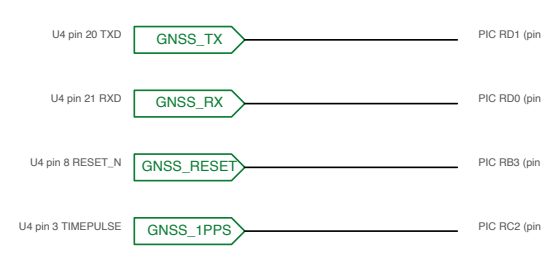
V_USB MUST BE TIED TO GND when USB is unused (IM 4.2.3, repeated in the 4.3 minimal design and the 4.9.1 checklist). This is not covered by the "inputs may be left open" rule.
V_BCKP maintains the RTC and battery-backed RAM, which is the difference between a hot start of a couple of seconds and a 24-29 s cold start every time the car is keyed off in the paddock. Backup draw is only 45 uA typ.
If no backup source is fitted, TIE V_BCKP DIRECTLY TO VCC - never leave it floating. Keep the line low-resistance: the switchover from main to backup draws a short current peak, and a resistive path turns that into a damaging voltage drop, so no series resistor "for isolation" here.

(4) RF FRONT END - one layout, passive or active antenna by populate / no-fit


VCC_RF
Bias-T, ACTIVE antenna only.
Fit L_bias / C_bias / R_bias and omit R_bias0;
for a passive antenna do exactly the reverse.
R_bias
C_bias
J_gnss
U.FL / SMA
RF
SHIELD
D_rf
L_bias
R_bias0
RF_IN
to U4 pin 11

The module has an internal DC block and internal 50 ohm matching, plus a SAW filter ahead of its LNA and an LTE band 13 notch. DO NOT FIT A DC BLOCK OR A MATCHING NETWORK on RF_IN - they are already inside, and the SAW is what lets an inexpensive passive antenna work next to a BG95 transmitting at ~22 dBm from a few tens of cm away.
R_bias is 22 ohm, NOT the 10 ohm in IM Table 26: supplying from VCC_RF at 3.3 V requires >= 19 ohm to hold short-circuit current under 150 mA, and 22 ohm is the nearest standard value. A chafed coax against bodywork is the most likely short in a race car, and this resistor is what stops it destroying the bias-T or the internal feed.
D_rf is REQUIRED (IM 4.5.2): an external RF connector is a conduction path for destructive signals. RF_IN routes as a grounded co-planar waveguide referenced to layer 2, matched to 50 ohm, via-shielded along its entire length, with vias around the RF_IN pad and NO STUBS. Absolute limits: 0 dBm in band, +13 dBm out of band, 30 dB max external gain - do not stack an active antenna with a further LNA.

(5) UART AND CONTROL TO THE PIC - direct, no level shifting, no series resistors


U4 pin 20 TXD
GNSS_TX
PIC RD1 (pin 43), U2RX
U4 pin 21 RXD
GNSS_RX
PIC RD0 (pin 42), U2TX
U4 pin 8 RESET_N
GNSS_RESET
PIC RB3 (pin 11), open-drain
U4 pin 3 TIMEPULSE
GNSS_1PPS
PIC RC2 (pin 40), CCP1 capture

Both ends sit on 3V3_SYS, so the levels are comfortable in both directions: the NEO needs Vih >= 0.8 x VCC = 2.64 V and swings to within 0.4 V of the rails at 2 mA, and the PIC is on the same rail. Max 5 mA per digital I/O, and NO HARDWARE FLOW CONTROL is supported. TIMEPULSE is worth routing even though the firmware does not use it yet - 30 ns RMS in hardware capture beats UART arrival time by orders of magnitude for correlating position to lap timing.